HACKATHON PROGRESS REPORT · 25 OCTOBER 2026



Local Agentic RAG for PDFs

Cited answers from any PDF, in English, Hindi, Hinglish and Marathi, using open-weight models on a free Google Colab T4 GPU

Satyam Kumar Pandey open-source project

About 80–85% complete · final by 31 Oct 2026

RESULTS AT A GLANCE (MEASURED ON A FRESH COLAB T4 RUNTIME)

73.1%
answer accuracy
61 English questions on 3 PDFs, up from 50.0% at baseline

92.3%
page recall@5
evidence page in the top 5, up from 73.1% for the starting code

88.9%
multilingual accuracy
20 questions: Hindi, Hinglish, Marathi, English, scanned pages

8.9 GB
GPU memory in use
of 15 GB on the T4, with the LLM, embedder and reranker loaded

1 The problem and the goal

Long PDFs such as research papers, government statistical releases and standards are hard to search. Hosted chatbots can answer questions about them, but they cost money, send the document to a third party, and often answer confidently when the PDF does not say. Our goal is an open-source chatbot that answers questions about *any* uploaded PDF under four constraints:

- **Local only.** Open-weight models and no API keys, so the document never leaves the runtime.
- **Free hardware.** One Colab T4: 15 GB GPU memory, 12.7 GB host RAM, 2 vCPUs.
- **Grounded.** Every answer cites the pages it used, or says the document does not contain the answer.
- **Multilingual.** Questions and documents may be in different languages, and scanned pages must work.

We started from the reference notebook and an early codebase that had only a non-agentic baseline (MiniLM search, no tables, no OCR, English only). We rebuilt ingestion, retrieval and the agent, and measured every change on a fixed benchmark.

WORKING TODAY

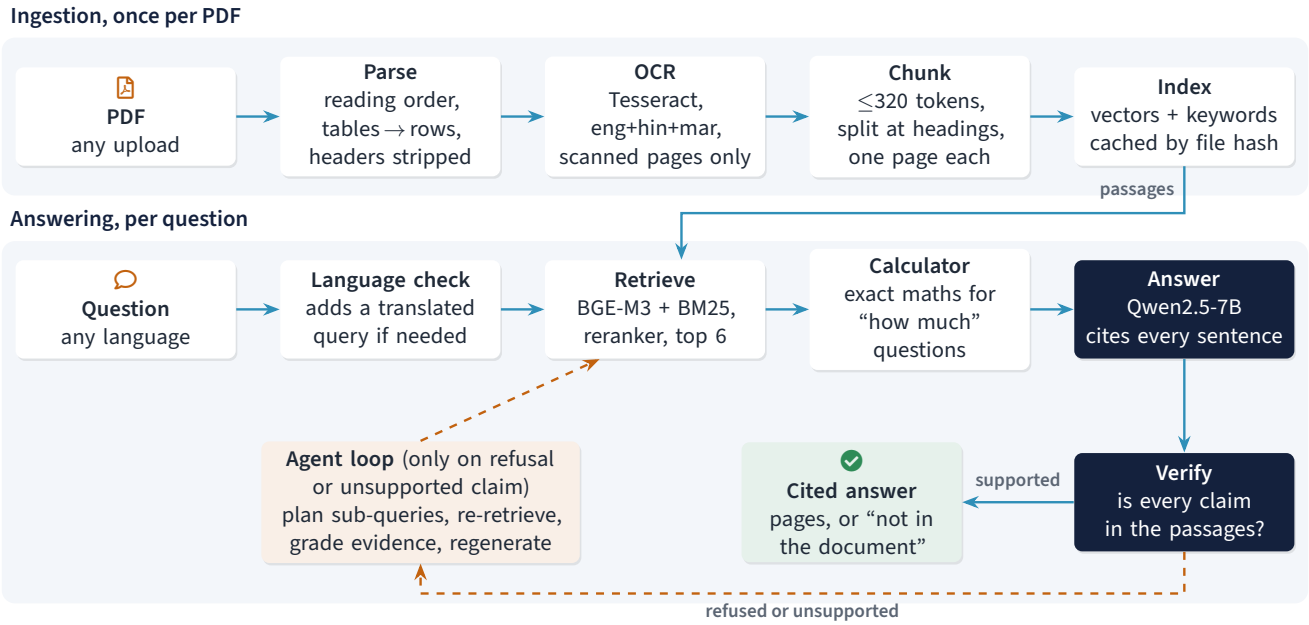
- Colab notebook that runs top to bottom on a fresh T4
- Chat UI (Gradio) with PDF upload, history and citations
- Tables, two-column layouts and scanned pages (OCR)
- 430 offline tests, about 15 seconds, no downloads

STILL OPEN

- Numerical questions on dense tables
- Larger, multi-annotator benchmark and an LLM judge
- Streaming answers in the chat UI
- Public release with an “Open in Colab” link

2 Approach and architecture

The system has two paths. **Ingestion** runs once per PDF and is cached by file hash. **Answering** runs per question. A single pass that retrieves, answers and verifies handles most questions, and the agent loop starts only when that pass refuses or the verifier finds an unsupported claim.



Navy boxes are LLM calls (Qwen2.5-7B-Instruct, 4-bit, served by Ollama). A typical question uses two to four calls, and the loop never exceeds eight. Structured steps use JSON-schema constrained output, and every step falls back to the simpler path if it fails.

Design decisions and the evidence behind them

Decision	Why	Measured effect
Hybrid search + cross-encoder reranker	Dense search finds paraphrases, BM25 finds exact labels and numbers, and the reranker reads question and passage together.	Page recall@5 73.1% → 96.2% (English models)
Table rebuilding	Statistical tables are aligned with spaces, so plain text extraction scrambles rows and columns.	Rows read as `label | 2025 Q2 = 3.8`. Table questions rose from 42% to 75%, helped by better retrieval
BGE-M3 + multilingual reranker	English-only models miss Hindi and Hinglish wording.	Multilingual set: recall@5 0.94 → 1.00 and MRR 0.88 → 0.97, with answer accuracy tied
Adaptive agent, not an always-on loop	Our first always-on loop was *worse* than a single pass (23 vs 27 of 38): sub-queries crowded out the real evidence.	The loop runs only when needed, with fewer false refusals at similar accuracy
Python calculator	The 7B model subtracts and counts wrong even with the right numbers in view.	Arithmetic is exact, but choosing the right table cell is still hard
Tesseract CLI OCR at 300 dpi	The built-in OCR path lost about 30% of the Hindi words on scanned pages.	Hindi word recall 0.70 → 0.99 and 1.00 on the two scanned pages
Ollama for the LLM	Prebuilt CUDA binaries. Source-built alternatives failed on Colab's CUDA 13 and Python 3.13.	Installs in about 40 s, and the 7B model needs 5.2 GB

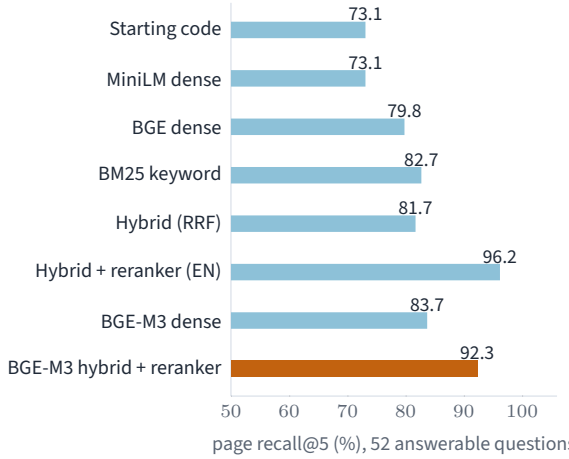
Engineering for a free runtime

The real limit was host RAM, not the GPU: the Ollama runner grew by about 100 MB per request until Colab killed the session. Memory-mapped weights, one request slot, retries and a watchdog that restarts the runner between requests fixed it. In the final run the watchdog restarted the runner and no question failed. Parsed documents are cached by file hash, so re-runs skip parsing.

3 Benchmarks

English set: 61 hand-written questions over three public PDFs: an ACL 2025 paper (17 pages, two columns), a U.S. BEA GDP release (28 pages, mostly statistical tables) and NIST AI 600-1 (64 pages). The questions cover seven categories, including 9 that the documents cannot answer. A correct answer must contain every required fact and must not refuse. **Multilingual set:** 20 questions over a Hindi article, a Marathi article, a scanned (image-only) Hindi document, and the English paper asked in Hindi and Hinglish. All results come from a fresh Colab T4 runtime.

Retrieval: is the evidence page in the top 5?



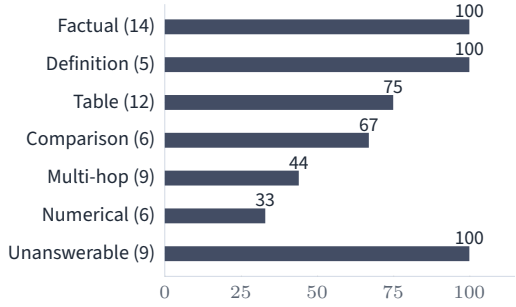
Orange is the default (multilingual). The reranker is the largest single gain. The multilingual default is one question below the English-only reranker here and ranks best at top-1 (66.3%).

Answers: end to end on the T4

	Baseline	Single pass	Agent
Answer accuracy	50.0%	69.2%	**73.1%**
Fact recall	53.2%	73.7%	77.2%
False refusals	26.9%	3.8%	1.9%
Unanswerable refused	9/9	9/9	9/9
Citation precision	70.6%	69.3%	73.5%
Latency, median	2.7 s	2.8 s	6.4 s

Same 61 questions and LLM in every column. “Baseline” is the starting design (MiniLM search, single pass), measured in an earlier T4 run. “Single pass” and “Agent” use the multilingual retriever. “Agent” is the default (adaptive loop + calculator). 95% intervals are about ±12 points, so small differences between the last two columns are within noise.

Agent accuracy by question type



Multilingual and scanned pages

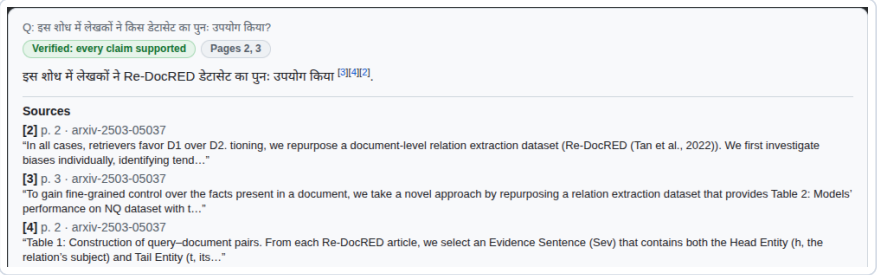
Question slice	n	Correct
Hindi questions (8 answerable)	9	8/8
of which on the scanned document (OCR)	2	2/2
Hinglish questions	3	2/3
Marathi questions (3 answerable)	4	3/3
English questions about Hindi/Marathi docs	4	3/4
Unanswerable, correctly refused	2	2/2
All answerable	18	**16/18 (88.9%)**

Answers come back in the question’s language. The two misses: a Hinglish two-part question that was refused, and “16 jets × 16 petals” computed as a sum. The same agent with English-only models also scores 16/18, but finds evidence less well (recall@5 0.94 vs 1.00).

Resources on the free T4

GPU memory in use, all three models loaded	8.9 / 15.0 GB
Ollama install + 4.7 GB model download	3.1 min
Ingest a 17-page paper (first time, cached after)	12.5 s
Scanned 2-page Hindi document, with OCR	41 s
Retrieval ablation, 8 configurations	4.0 min
English answers, 2 configurations × 61	15.0 min
Multilingual answers, 2 configurations × 20	7.6 min
Offline test suite	430 tests, ≈15 s

What an answer looks like


Q: इस शोध में लेखकों ने किस डेटासेट का पुनः उपयोग किया?

Verified: every claim supported Pages 2, 3

इस शोध में लेखकों ने Re-DocRED डेटासेट का पुनः उपयोग किया [3][4][2].

Sources

[2] p. 2 · arxiv-2503-05037
“In all cases, retrievers favor D1 over D2. tioning, we repurpose a document-level relation extraction dataset (Re-DocRED (Tan et al., 2022)). We first investigate biases individually, identifying tend…”

[3] p. 3 · arxiv-2503-05037
“To gain fine-grained control over the facts present in a document, we take a novel approach by repurposing a relation extraction dataset that provides Table 2: Models’ performance on NQ dataset with t…”

[4] p. 2 · arxiv-2503-05037
“Table 1: Construction of query–document pairs. From each Re-DocRED article, we select an Evidence Sentence (Sev) that contains both the Head Entity (h, the relation’s subject) and Tail Entity (t, its…”

Screenshot from the T4 run. A Hindi question about the English paper (“Which dataset did the authors repurpose?”) is answered in Hindi. The answer names Re-DocRED, cites pages 2 and 3, and the verifier marks every claim as supported. The quoted sources are the passages the citations point to.

4 Progress

About 80–85% of the planned scope is built and measured — **final: 31 Oct**

- ✅ PDF ingestion: reading order, tables, headers, OCR
- ✅ Hybrid retrieval with reranking and an index cache
- ✅ Adaptive agent: plan, grade, verify, calculator
- ✅ Page citations, and refusals when the PDF has no answer
- ✅ Hindi, Hinglish, Marathi, cross-language questions
- ✅ Colab notebook and chat UI, run on a fresh T4
- ✅ Benchmarks with ablations and 430 tests
- 🕒 Numerical questions: pick the right table cells
- 🕒 Larger benchmark, second annotator, LLM judge
- 🕒 More Marathi, English-to-Indic and scanned questions
- 🕒 Streaming answers in the chat UI
- 🕒 Public repository with an “Open in Colab” link

Latest improvements: before and after

We analysed every failed question from the previous full run, fixed only what the evidence pointed to, added regression tests for each fix, and re-ran all benchmarks in a fresh runtime.

Change	Evidence that it was needed	Before → after
OCR through the Tesseract CLI	One of the two scanned-page questions failed because the OCR text had lost words	Word recall 0.70 → 0.99 and 1.00. Scanned-page questions 1/2 → 2/2
Refusal detection	An unanswerable question got a cited “the document does not say…” that was scored as an answer	Unanswerable refused 8/9 → 9/9
Calculator with fixed operations	It returned nothing for 16×16 and some differences, and the model ignored its result	Numerical questions 1/6 → 2/6 for the agent. One question is within noise
Table rows and headers	Stacked cells and spanning headers gave wrong values on two BEA tables	Correct values on the tables we checked, but one BEA question now ranks lower (recall@5 94.2% → 92.3%)
Shared parsing across variants	The benchmark parsed every PDF once per configuration	Benchmark time 31 → 27 min (faster answers also helped)

One change made results worse and was reverted: a stricter “refuse unless the exact fact is present” prompt rule raised false refusals from 3.8% to 21.2% in a single-pass run. Net effect on the English set: answer accuracy is unchanged at 73.1% (numerical +1, unanswerable +1, multi-hop −1).

5 Limitations

- **Small benchmarks.** 61 English and 20 multilingual questions, written by one person. Grading matches required facts, which can miss a correct paraphrase. We report 95% intervals for this reason.
- **Numerical questions** remain the weakest category. Retrieval finds the right page, but the 7B model often picks a number from the wrong row or column of a dense table.
- **Language coverage** is what BGE-M3 and Qwen2.5-7B handle well. We measured Hindi, Hinglish and Marathi only, on 20 questions. We do not claim support for other languages.
- **Not interpreted:** charts and figures. OCR quality depends on the scan.

6 What we learned

- **Retrieval moves accuracy the most.** Better search and reranking added 21 points of answer accuracy (50% → 71%). The agent loop and tools added a few more, mostly by refusing less often.
- **Measure on the real target.** The planning documents assumed CUDA 12 and Python 3.10 to 3.12, but Colab now has CUDA 13 and Python 3.13. Host RAM, not the GPU, was the limit that crashed runs.
- **Every change needs a benchmark run.** A prompt rule that looked safe raised false refusals from 3.8% to 21.2%. Only the benchmark showed it.

7 Next steps

The remaining work will be completed before 31 October 2026.